

4.10.12

EE25BTECH11011 - Navya

Question:

The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Solution:

The equations of the given lines are

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 10 \quad (1)$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \mathbf{x} = -5 \quad (2)$$

Write the system as a single matrix equation:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \end{pmatrix} \quad (3)$$

Solve using the inverse:

$$\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 10 \\ -5 \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ -5 \end{pmatrix} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \quad (4)$$

Thus, the intersection point is

$$\mathbf{x} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \quad (5)$$

Now, check whether it lies on the line

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \mathbf{x} = 0 \quad (6)$$

Substituting,

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} = \frac{1}{3}(-100 + 100) = 0 \quad (7)$$

Hence, The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

